

LINEAR TIME-INVARIANT SYSTEMS AND THE MATRIX EXPONENTIAL

1. THE SETUP

A (continuous-time) linear system with constant coefficients has the form

$$\dot{x}(t) = A x(t) + f(t), \quad x(t) \in \mathbb{R}^n, \quad A \in \mathbb{R}^{n \times n}.$$

Homogeneous: $f(t) \equiv 0$ so $\dot{x} = Ax$. **Nonhomogeneous:** $f(t)$ is a given input/forcing. For any initial condition $x(t_0) = x_0$, existence and uniqueness hold and the solution is smooth.

2. THE MATRIX EXPONENTIAL e^{At}

Definition (power series).

$$e^{At} = \sum_{k=0}^{\infty} \frac{(At)^k}{k!} = I + At + \frac{(At)^2}{2!} + \dots$$

This converges for all $t \in \mathbb{R}$ and any square A .

Characterization. $Y(t) = e^{At}$ is the unique matrix function satisfying

$$\frac{d}{dt} Y(t) = A Y(t), \quad Y(0) = I.$$

Proposition 2.1 (Matrix exponential solves $Y' = AY$). *Let $A \in \mathbb{C}^{n \times n}$ and define*

$$E(t) := e^{tA} := \sum_{k=0}^{\infty} \frac{(At)^k}{k!} \quad (t \in \mathbb{R}).$$

Then the series converges for all t , E is differentiable on $\mathbb{R}$, and

$$E'(t) = A E(t) = E(t) A, \quad E(0) = I.$$

In particular, E is the unique solution of the matrix IVP $Y'(t) = A Y(t)$, $Y(0) = I$.

Proof. Pick any submultiplicative matrix norm $\|\cdot\|$. Then

$$\sum_{k=0}^{\infty} \left\| \frac{(At)^k}{k!} \right\| \leq \sum_{k=0}^{\infty} \frac{(|t| \|A\|)^k}{k!} = e^{|t| \|A\|} < \infty,$$

so the series converges absolutely for each t , and uniformly on compact t -intervals. Similarly,

$$\sum_{k=1}^{\infty} \left\| \frac{A(At)^{k-1}}{(k-1)!} \right\| \leq \|A\| e^{|t| \|A\|},$$

so the series of termwise derivatives also converges uniformly on compacts. Hence differentiation term-by-term is justified:

$$E'(t) = \sum_{k=1}^{\infty} \frac{A(At)^{k-1}}{(k-1)!} = A \sum_{k=0}^{\infty} \frac{(At)^k}{k!} = A E(t).$$

Evaluating at $t = 0$ gives $E(0) = I$. Uniqueness of solutions to $Y' = AY$, $Y(0) = I$ follows from linear ODE theory (or by noting any two solutions Y_1, Y_2 satisfy $(Y_1 - Y_2)' = A(Y_1 - Y_2)$ with zero initial data, hence $Y_1 \equiv Y_2$). $\square$

Key properties.

- $\frac{d}{dt} e^{At} = A e^{At} = e^{At} A$.
- **Semigroup:** $e^{A(t+s)} = e^{At} e^{As}$.
- **Invertibility:** $(e^{At})^{-1} = e^{-At}$.
- **Change of coordinates:** if S is invertible, then $e^{S^{-1}AS t} = S^{-1} e^{At} S$.
- **Determinant/trace link:** $\det(e^{At}) = e^{(\text{tr } A)t}$.

3. SOLVING THE HOMOGENEOUS IVP $\dot{x} = Ax$

For $x(t_0) = x_0$,

$$x(t) = e^{A(t-t_0)} x_0.$$

The columns of e^{At} form a fundamental matrix of solutions; any solution is a linear combination of those columns.

4. COMPUTING e^{At} IN PRACTICE

One typically does not compute the power series directly. Common methods:

(a) Diagonalization. If $A = V \Lambda V^{-1}$ with $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$, then

$$e^{At} = V e^{\Lambda t} V^{-1}, \quad e^{\Lambda t} = \text{diag}(e^{\lambda_1 t}, \dots, e^{\lambda_n t}).$$

(b) Jordan form. If $A = V J V^{-1}$ with a Jordan block $J_k(\lambda) = \lambda I + N$ where N is nilpotent (ones on the superdiagonal), then

$$e^{J_k(\lambda)t} = e^{\lambda t} \left(I + Nt + \frac{N^2 t^2}{2!} + \dots + \frac{N^{k-1} t^{k-1}}{(k-1)!} \right), \quad e^{At} = V e^{Jt} V^{-1}.$$

(c) Triangular & 2×2 shortcuts. If A is upper triangular, e^{At} is upper triangular with (i, i) -entry $e^{a_{ii}t}$. For $A = \begin{bmatrix} a & b \\ 0 & d \end{bmatrix}$,

$$e^{At} = \begin{bmatrix} e^{at} & b \frac{e^{at} - e^{dt}}{a - d} \\ 0 & e^{dt} \end{bmatrix} \quad (a \neq d), \quad \begin{bmatrix} e^{at} & b t e^{at} \\ 0 & e^{at} \end{bmatrix} \quad (a = d).$$

(d) 2×2 trace-determinant formula. Let $\mu = \frac{1}{2} \text{tr}(A)$ and $\delta = \det(A)$, and set $\omega = \sqrt{\mu^2 - \delta}$. Then

$$e^{At} = e^{\mu t} \left(\cosh(\omega t) I + \frac{\sinh(\omega t)}{\omega} (A - \mu I) \right) \quad (\mu^2 \geq \delta).$$

If $\mu^2 < \delta$, put $\omega = i\beta$ with $\beta = \sqrt{\delta - \mu^2}$ to get the equivalent real form

$$e^{At} = e^{\mu t} \left(\cos(\beta t) I + \frac{\sin(\beta t)}{\beta} (A - \mu I) \right).$$

(e) Laplace transform viewpoint.

$$\mathcal{L}\{e^{At}\}(s) = (sI - A)^{-1},$$

so partial fractions of $(sI - A)^{-1}$ invert to give e^{At} .

Numerical note: reliable software computes e^{At} via scaling & squaring with Padé approximants; diagonalizing by hand can be ill-conditioned near defective matrices.

5. FROM COMPLEX EIGENVALUES TO REAL SOLUTIONS

If A has complex eigenvalues $\lambda = \alpha \pm i\beta$ ($\beta \neq 0$), solutions contain factors $e^{\alpha t} \cos(\beta t)$ and $e^{\alpha t} \sin(\beta t)$. Real and imaginary parts of $e^{(\alpha+i\beta)t} v$ (for a complex eigenvector v) give a real basis of solutions.

6. THE NONHOMOGENEOUS SYSTEM $\dot{x} = Ax + f(t)$

Proposition 6.1 (Duhamel / Variation of constants). *Let $A \in \mathbb{R}^{n \times n}$ be constant, $t_0 \in \mathbb{R}$, $x_0 \in \mathbb{R}^n$, and $f \in L^1_{\text{loc}}(\mathbb{R}; \mathbb{R}^n)$. Then the unique solution of*

$$\dot{x}(t) = Ax(t) + f(t), \quad x(t_0) = x_0,$$

is

$$x(t) = e^{A(t-t_0)}x_0 + \int_{t_0}^t e^{A(t-s)}f(s) ds.$$

Moreover $x \in AC_{\text{loc}}$, $x'(t) = Ax(t) + f(t)$ for a.e. t , and if $f \in C^k$ then $x \in C^{k+1}$.

Proof 1 (integrating factor, first principles). Fix a compact interval $I = [a, b] \ni t_0$. The map $t \mapsto e^{-At}$ is C^1 and bounded on I . If x solves $\dot{x} = Ax + f$ in the a.e. sense with $x \in AC(I)$, then the product rule for AC functions gives

$$\frac{d}{dt}(e^{-At}x(t)) = (-Ae^{-At})x(t) + e^{-At}\dot{x}(t) = e^{-At}f(t) \quad \text{a.e. on } I.$$

Integrate from t_0 to $t \in I$:

$$e^{-At}x(t) - e^{-At_0}x(t_0) = \int_{t_0}^t e^{-As}f(s) ds.$$

Multiplying by e^{At} on the left and using $x(t_0) = x_0$ yields the formula.

Conversely, define x by the displayed formula. Then x is absolutely continuous on I (Volterra integral with bounded kernel against L^1 data). Differentiate under the integral sign (dominated convergence using $\|Ae^{A(t-s)}\| \leq C$ on $I \times I$) and use Leibniz' rule:

$$\frac{d}{dt} \left(\int_{t_0}^t e^{A(t-s)}f(s) ds \right) = e^{A(t-t)}f(t) + \int_{t_0}^t Ae^{A(t-s)}f(s) ds = f(t) + \int_{t_0}^t Ae^{A(t-s)}f(s) ds.$$

Hence $x'(t) = Ae^{A(t-t_0)}x_0 + f(t) + \int_{t_0}^t Ae^{A(t-s)}f(s) ds = Ax(t) + f(t)$ a.e., and $x(t_0) = x_0$ is immediate. Uniqueness follows by applying this to the difference of two solutions (or by Grönwall). $\square$

Proof 2 (Laplace transform; assumes $t_0 = 0$ and f of exponential order). For simplicity take $t_0 = 0$ and extend by shift otherwise. Let $\mathcal{L}\{x\}(s) = X(s)$. Taking Laplace transforms of $\dot{x} = Ax + f$ with $x(0) = x_0$ gives

$$sX(s) - x_0 = AX(s) + F(s) \quad \Rightarrow \quad X(s) = (sI - A)^{-1}(x_0 + F(s)).$$

Since $\mathcal{L}\{e^{At}\}(s) = (sI - A)^{-1}$, inversion yields

$$x(t) = e^{At}x_0 + \int_0^t e^{A(t-s)}f(s) ds.$$

$\square$

Remark 6.2 (Semigroup form and minimal hypotheses). Writing $T(t) := e^{At}$, Duhamel is $x(t) = T(t-t_0)x_0 + \int_{t_0}^t T(t-s)f(s) ds$. The same proof works in any Banach space for a C^0 -semigroup $(T(t))_{t \geq 0}$ (Hille–Yosida theory), with $f \in L^1_{\text{loc}}$ giving a *mild* solution; if f is continuous, the solution is classical.

6.1. Regularity from Duhamel's formula.

Proposition 6.3 (First-principles $C^k \rightarrow C^{k+1}$). *Fix $A \in \mathbb{R}^{n \times n}$, $t_0 \in \mathbb{R}$, $x_0 \in \mathbb{R}^n$, and let $I = [a, b]$ with $t_0 \in I$. For $f \in C^k(I; \mathbb{R}^n)$ ($k \in \mathbb{N} \cup \{0\}$) define*

$$x(t) := e^{A(t-t_0)}x_0 + \int_{t_0}^t e^{A(t-s)} f(s) ds, \quad t \in I.$$

Then $x \in C^{k+1}(I)$ and, for $0 \leq m \leq k$,

$$x^{(m+1)}(t) = A x^{(m)}(t) + f^{(m)}(t) \quad (t \in I).$$

Proof from first principles. Step 1: The case $k = 0$ ($f \in C^0$) gives $x \in C^1$ and $x' = Ax + f$. Write $x(t) = E(t) + F(t)$ with $E(t) := e^{A(t-t_0)}x_0$ and $F(t) := \int_{t_0}^t e^{A(t-s)} f(s) ds$. Since $t \mapsto e^{At}$ is C^1 and $(e^{At})' = Ae^{At}$, we have $E'(t) = AE(t)$.

For F , fix $t \in (a, b)$ and $h \rightarrow 0$. Then

$$\frac{F(t+h) - F(t)}{h} = \frac{1}{h} \int_t^{t+h} e^{A(t+h-s)} f(s) ds + \int_{t_0}^t \frac{e^{A(t+h-s)} - e^{A(t-s)}}{h} f(s) ds =: I_1(h) + I_2(h).$$

For $I_1(h)$, the continuity of $(t, s) \mapsto e^{A(t-s)} f(s)$ implies

$$\lim_{h \rightarrow 0} I_1(h) = \lim_{h \rightarrow 0} \frac{1}{h} \int_t^{t+h} e^{A(t-s)} f(s) ds = e^{A \cdot 0} f(t) = f(t),$$

by the fundamental theorem of calculus for averages of a continuous function.

For $I_2(h)$, use the mean value (integral) form of the increment:

$$\frac{e^{A(t+h-s)} - e^{A(t-s)}}{h} = \int_0^1 A e^{A(t-s+\theta h)} d\theta,$$

hence

$$I_2(h) = \int_{t_0}^t \left(\int_0^1 A e^{A(t-s+\theta h)} d\theta \right) f(s) ds.$$

On I the integrand is uniformly bounded; by continuity in (θ, s) we may pass to the limit under the integrals to get

$$\lim_{h \rightarrow 0} I_2(h) = \int_{t_0}^t A e^{A(t-s)} f(s) ds.$$

Therefore $F'(t) = f(t) + \int_{t_0}^t A e^{A(t-s)} f(s) ds$, and

$$x'(t) = E'(t) + F'(t) = AE(t) + \int_{t_0}^t A e^{A(t-s)} f(s) ds + f(t) = Ax(t) + f(t).$$

The right-hand side is continuous on I , so $x \in C^1(I)$.

Step 2: Induction. Suppose $f \in C^k$ with $k \geq 1$ and the claim holds up to some $m < k$. From Step 1, $x \in C^1$ and $x' = Ax + f$. Differentiate this identity m times (ordinary product rule; A is constant):

$$x^{(m+1)} = A x^{(m)} + f^{(m)}.$$

By the inductive hypothesis, $x^{(m)}$ is continuous and $f^{(m)} \in C^{k-m}$, so $x^{(m+1)}$ is continuous. This closes the induction up to $m = k$, yielding $x \in C^{k+1}$ and the stated formula. $\square$

Let $\Omega \subset \mathbb{R}^n$ be bounded and let $0 < \alpha \leq 1$. For a bounded function $f : \Omega \rightarrow \mathbb{R}$, define the Hölder (semi)norm

$$[f]_{C^{0,\alpha}(\Omega)} := \sup_{\substack{x,y \in \Omega \\ x \neq y}} \frac{|f(x) - f(y)|}{|x - y|^\alpha}.$$

The full $C^{0,\alpha}$ norm is

$$\|f\|_{C^{0,\alpha}(\Omega)} := \sup_{x \in \Omega} |f(x)| + [f]_{C^{0,\alpha}(\Omega)}.$$

For $k \in \mathbb{N}$ and a multi-index β (with $|\beta|$ its order), write $D^\beta f$ for the corresponding partial derivative. The $C^{k,\alpha}$ norm is

$$\|f\|_{C^{k,\alpha}(\Omega)} := \sum_{|\beta| \leq k} \sup_{x \in \Omega} |D^\beta f(x)| + \sum_{|\beta|=k} \sup_{\substack{x,y \in \Omega \\ x \neq y}} \frac{|D^\beta f(x) - D^\beta f(y)|}{|x - y|^\alpha}.$$

Notes. (1) $[f]_{C^{0,\alpha}(\Omega)} = 0$ for constant functions; the full norm adds the sup norm.

(2) The case $\alpha = 1$ recovers the Lipschitz seminorm.

(3) One often takes Ω open and interprets the suprema over the closure $\bar{\Omega}$.

Proposition 6.4 (First-principles Hölder gain: $f \in C^{0,\alpha} \Rightarrow x \in C^{1,\alpha}$). *Let $I = [a, b]$ and $f \in C^{0,\alpha}(I; \mathbb{R}^n)$ with $\alpha \in (0, 1]$. With x as above, $x \in C^{1,\alpha}(I)$ and there exists $C = C(A, I) > 0$ such that*

$$\|x'\|_{C^{0,\alpha}(I)} \leq C(\|x\|_{C^0(I)} + \|f\|_{C^{0,\alpha}(I)}).$$

Proof from first principles. From Proposition 6.3 with $k = 0$, $x \in C^1$ and

$$x'(t) = Ax(t) + f(t) = Ae^{A(t-t_0)}x_0 + \int_{t_0}^t Ae^{A(t-s)}f(s)ds + f(t).$$

Fix $t, u \in I$, WLOG $t > u$. Then

$$\begin{aligned} x'(t) - x'(u) &= (f(t) - f(u)) + \int_u^t Ae^{A(t-s)}f(s)ds + \int_a^u (Ae^{A(t-s)} - Ae^{A(u-s)})f(s)ds \\ &=: T_1 + T_2 + T_3. \end{aligned}$$

Estimate each term:

(1) *Hölder term.* Since $f \in C^{0,\alpha}$,

$$\|T_1\| \leq [f]_{C^{0,\alpha}(I)} |t - u|^\alpha.$$

(2) *Short-interval integral.* On I there is $M = \sup_{(r,s) \in I^2} \|Ae^{A(r-s)}\| < \infty$; hence

$$\|T_2\| \leq \int_u^t M \|f(s)\| ds \leq M \|f\|_{C^0(I)} |t - u|.$$

(3) *Kernel difference.* The map $r \mapsto Ae^{A(r-s)}$ is C^1 with derivative $A^2e^{A(r-s)}$; thus by the mean value theorem there is $L = \sup_{(r,s) \in I^2} \|A^2e^{A(r-s)}\|$ so that

$$\|Ae^{A(t-s)} - Ae^{A(u-s)}\| \leq L |t - u| \quad (s \in I).$$

Therefore

$$\|T_3\| \leq \int_a^u L |t - u| \|f(s)\| ds \leq L |t - u| \|f\|_{C^0(I)} (u - a).$$

Combining,

$$\|x'(t) - x'(u)\| \leq [f]_{C^{0,\alpha}} |t - u|^\alpha + C_1 |t - u|,$$

where $C_1 = C_1(A, I) \|f\|_{C^0(I)}$. Since $|t - u| \leq (b - a)^{1-\alpha} |t - u|^\alpha$ for $t, u \in I$, we get

$$\|x'(t) - x'(u)\| \leq ([f]_{C^{0,\alpha}} + C_1 (b - a)^{1-\alpha}) |t - u|^\alpha,$$

which proves $x' \in C^{0,\alpha}(I)$ with the stated bound (also using $\|x'\|_{C^0} \leq \|A\| \|x\|_{C^0} + \|f\|_{C^0}$). $\square$

Proposition 6.5 (Regularity from Duhamel's formula). *Fix $A \in \mathbb{R}^{n \times n}$, $t_0 \in \mathbb{R}$, $x_0 \in \mathbb{R}^n$, and set*

$$x(t) := e^{A(t-t_0)}x_0 + \int_{t_0}^t e^{A(t-s)}f(s)ds \quad (t \in \mathbb{R}),$$

whenever the integral makes sense. Then on every compact interval $I \subset \mathbb{R}$ the following hold.

(i) *If $f \in L^1_{\text{loc}}(\mathbb{R}; \mathbb{R}^n)$, then $x \in AC(I)$ and*

$$x'(t) = Ax(t) + f(t) \quad \text{for a.e. } t \in I.$$

In particular x is the unique solution of $\dot{x} = Ax + f$ with $x(t_0) = x_0$.

(ii) If $f \in L^p_{\text{loc}}(\mathbb{R}; \mathbb{R}^n)$ for some $1 \leq p \leq \infty$, then $x \in W^{1,p}(I)$ and

$$x' = Ax + f \text{ in } L^p(I).$$

(iii) If f is piecewise C^r on I with jump discontinuities at $\{\tau_j\} \subset I$, then $x \in C^{r+1}$ on each open subinterval between the τ_j , x is continuous on I , and

$$x'_+(\tau_j) - x'_-(\tau_j) = f_+(\tau_j) - f_-(\tau_j).$$

Proof. Fix a compact $I = [a, b] \subset \mathbb{R}$. The kernel $K(t, s) := e^{A(t-s)}$ is smooth in t , jointly continuous on $\{(t, s) \in I \times [a, b] : s \leq t\}$, and there exist constants M, M_1 (depending on A and I) with

$$\|K(t, s)\| \leq M, \quad \|\partial_t K(t, s)\| = \|Ae^{A(t-s)}\| \leq M_1 \quad (a \leq s \leq t \leq b).$$

(i) If $f \in L^1_{\text{loc}}$, then $F(t) := \int_{t_0}^t K(t, s)f(s) ds$ is absolutely continuous on I (Volterra integral with bounded kernel). By differentiation under the integral sign (dominated convergence using $M_1\|f\| \in L^1$) and Leibniz' rule,

$$F'(t) = K(t, t)f(t) + \int_{t_0}^t \partial_t K(t, s)f(s) ds = f(t) + \int_{t_0}^t Ae^{A(t-s)}f(s) ds.$$

Hence $x' = Ae^{A(t-t_0)}x_0 + F'(t) = Ax(t) + f(t)$ a.e. Uniqueness for $\dot{x} = Ax + f$ follows from linearity (apply Grönwall to the difference).

(ii) Since $x' = Ax + f$ a.e. on I and $x \in L^\infty(I)$ (bounded kernel against L^1 on I), we get $x' \in L^p(I)$ whenever $f \in L^p(I)$.

(iii) On each subinterval of smoothness, apply (iii). Continuity of x at a jump τ follows directly from the integral representation. The jump of x' equals the jump of f because $x' = Ax + f$ and x is continuous. $\square$

Two useful special cases:

- **Constant input** $f(t) \equiv b$:

$$x(t) = e^{A(t-t_0)}x_0 + \left(\int_{t_0}^t e^{A(t-s)} ds \right) b.$$

If A is invertible,

$$\int_{t_0}^t e^{A(t-s)} ds = A^{-1} (e^{A(t-t_0)} - I),$$

so

$$x(t) = e^{A(t-t_0)}x_0 + A^{-1} (e^{A(t-t_0)} - I) b.$$

The equilibrium (steady state) x^* solves $Ax^* + b = 0$, i.e. $x^* = -A^{-1}b$ if A is invertible. Then

$$x(t) = x^* + e^{A(t-t_0)}(x_0 - x^*).$$

- **Impulse input:** δ -like inputs are propagated by e^{At} (the *state-transition matrix*).

7. LONG-TIME BEHAVIOR AND STABILITY

Let the *spectral abscissa* be $\alpha(A) = \max_i \Re(\lambda_i)$.

- If $\alpha(A) < 0$: $e^{At} \rightarrow 0$ as $t \rightarrow \infty$ (asymptotically stable).
- If $\alpha(A) > 0$: solutions grow exponentially (unstable).
- If $\alpha(A) = 0$: marginal/critical—growth can be polynomial times oscillatory (e.g. Jordan blocks at $\Re \lambda = 0$), see Proposition 7.4.

Proposition 7.1 (Jordan-block estimate and growth trichotomy). *Let $A \in \mathbb{C}^{n \times n}$ and write its Jordan decomposition $A = VJV^{-1}$, where $J = \text{diag}(J_{k_1}(\lambda_1), \dots, J_{k_s}(\lambda_s))$ with $J_k(\lambda) = \lambda I + N$ and N the nilpotent Jordan off-diagonal (so $N^k = 0$). Set the spectral abscissa $\alpha(A) := \max_i \Re \lambda_i$, and let*

$$m := \max\{k_j - 1 : \Re \lambda_j = \alpha(A)\} \quad (\text{take } m = 0 \text{ if no eigenvalue has } \Re \lambda = \alpha(A)).$$

Then there exists a constant $K \geq 1$ (depending on A and the chosen operator norm) such that, for all $t \geq 0$,

$$\|e^{At}\| \leq K e^{\alpha(A)t} (1+t)^m. \quad (1)$$

Moreover, if there is a Jordan block of size $k \geq 2$ with $\Re \lambda = \alpha(A)$, then there exists a nonzero vector v and constants $c > 0$, $t_0 \geq 0$ with

$$\|e^{At}v\| \geq c e^{\alpha(A)t} t^{k-1} \quad \text{for all } t \geq t_0. \quad (2)$$

Consequently:

- If $\alpha(A) < 0$, then $\|e^{At}\| \rightarrow 0$ as $t \rightarrow \infty$ (exponential decay).
- If $\alpha(A) = 0$, then $\|e^{At}\| \lesssim (1+t)^m$; in particular, $\sup_{t \geq 0} \|e^{At}\| < \infty$ iff every Jordan block with $\Re \lambda = 0$ has size 1 (semisimple on the imaginary axis).
- If $\alpha(A) > 0$, then along some $v \neq 0$, $\|e^{At}v\| \gtrsim e^{\alpha(A)t}$, and if a top block has size $k \geq 2$, the lower bound (2) holds with the polynomial factor t^{k-1} .

Proof. Step 1: Single Jordan block. Fix $k \geq 1$ and $\lambda \in \mathbb{C}$. Since $N^k = 0$,

$$e^{J_k(\lambda)t} = e^{(\lambda I + N)t} = e^{\lambda t} \sum_{r=0}^{k-1} \frac{(Nt)^r}{r!}.$$

Let $\|\cdot\|$ be any submultiplicative matrix norm. Because the finite set $\{N^r : 0 \leq r \leq k-1\}$ is bounded, there exists $C_k \geq 1$ with $\|N^r\| \leq C_k$ for all $r \leq k-1$. Hence, for all $t \geq 0$,

$$\|e^{J_k(\lambda)t}\| \leq e^{\Re \lambda t} \sum_{r=0}^{k-1} \frac{\|N^r\| t^r}{r!} \leq C_k e^{\Re \lambda t} \sum_{r=0}^{k-1} \frac{t^r}{r!} \leq C'_k e^{\Re \lambda t} (1+t)^{k-1}, \quad (3)$$

for a suitable $C'_k > 0$ (absorbing the factorials into the constant and the $(1+t)^{k-1}$ upper bound).

Step 2: Block diagonal J . Since J is block diagonal, e^{Jt} is block diagonal with diagonal blocks $e^{J_{k_j}(\lambda_j)t}$. Using (3) and that the norm of a block diagonal matrix is bounded by a constant times the maximum (or sum) of block norms, there exists $C \geq 1$ with

$$\|e^{Jt}\| \leq C \max_j \|e^{J_{k_j}(\lambda_j)t}\| \leq C e^{\alpha(A)t} (1+t)^m.$$

Conjugating by V ,

$$\|e^{At}\| = \|V e^{Jt} V^{-1}\| \leq \|V\| \|V^{-1}\| \|e^{Jt}\| \leq K e^{\alpha(A)t} (1+t)^m,$$

with $K := C \|V\| \|V^{-1}\|$. This proves (1).

Step 3: Lower bound when a top block is nontrivial. Assume J has a block $J_k(\lambda)$ with $\Re \lambda = \alpha(A)$ and $k \geq 2$. In the Jordan basis, let $e_1, \dots, e_k$ be the standard chain in that block (so $N e_j = e_{j-1}$, $e_0 = 0$). Then

$$e^{J_k(\lambda)t} e_k = e^{\lambda t} \sum_{r=0}^{k-1} \frac{t^r}{r!} N^r e_k = e^{\lambda t} \left(e_k + t e_{k-1} + \dots + \frac{t^{k-1}}{(k-1)!} e_1 \right).$$

Hence the e_1 -component equals $e^{\lambda t} \frac{t^{k-1}}{(k-1)!}$. Therefore, for large t ,

$$\|e^{Jt} w\| \geq c_0 e^{\alpha(A)t} t^{k-1}$$

for some w supported on that block (take $w = e_k$) and constant $c_0 > 0$ depending only on the norm and k . Returning to the original basis with $v := Vw \neq 0$, we get

$$\|e^{At}v\| = \|V e^{Jt} w\| \geq \|V^{-1}\|^{-1} \|e^{Jt} w\| \geq c e^{\alpha(A)t} t^{k-1}$$

for all $t \geq t_0$, with $c = c_0 / \|V^{-1}\|$ and some $t_0 \geq 0$. This is (2).

Consequences. If $\alpha(A) < 0$, (1) gives $\|e^{At}\| \rightarrow 0$. If $\alpha(A) = 0$, (1) yields a polynomial bound, and boundedness holds iff $m = 0$, i.e. every block with $\Re \lambda = 0$ has size 1. If $\alpha(A) > 0$, the lower bound above shows exponential (and possibly polynomially corrected) growth along some direction. $\square$

Remark 7.2 (Why the hypothesis “ $k \geq 2$ ”). The clause “there exists a Jordan block of size $k \geq 2$ with $\Re \lambda = \alpha(A)$ ” in the lower bound emphasizes the *polynomial* correction: it ensures the factor t^{k-1} in addition to the leading $e^{\alpha(A)t}$. Instability does *not* require $k \geq 2$: if $\alpha(A) > 0$ and there is an eigenvalue λ with $\Re \lambda = \alpha(A)$ and a (size-1) Jordan block, then for an eigenvector $v \neq 0$,

$$e^{At}v = e^{\lambda t}v \quad \Rightarrow \quad \|e^{At}v\| = e^{\alpha(A)t} \|v\|,$$

so solutions already grow exponentially. By contrast, in the critical case $\alpha(A) = 0$, size-1 Jordan blocks on the imaginary axis yield bounded (oscillatory) dynamics, while size ≥ 2 produce polynomial growth via the t^{k-1} factor.

Corollary 7.3 (Sharpened lower bound without $k \geq 2$). *If λ is an eigenvalue with $\Re \lambda = \alpha(A)$, then there exist $c > 0$, $t_0 \geq 0$ and $v \neq 0$ such that*

$$\|e^{At}v\| \geq c e^{\alpha(A)t} \quad \text{for all } t \geq t_0.$$

If, moreover, the Jordan block for λ has size $k \geq 2$, the stronger bound $\|e^{At}v\| \geq c e^{\alpha(A)t} t^{k-1}$ holds for some (possibly different) $v \neq 0$ and all $t \geq t_0$.

For $n = 2$, a quick classification using $\text{tr}(A)$ and $\det(A)$:

- $\det A < 0$: **saddle** (one eigenvalue > 0 , one < 0 , unstable).
- $\det A > 0$, $\text{tr } A < 0$: **stable** (node if discriminant ≥ 0 , spiral if < 0).
- $\det A > 0$, $\text{tr } A > 0$: **unstable** (node/spiral).
- $\det A > 0$, $\text{tr } A = 0$, discriminant < 0 : **center** (pure rotation; neutral stability).

Proposition 7.4 (Polynomial-times-oscillatory structure at the imaginary axis). *Let $A \in \mathbb{R}^{n \times n}$ (or $\mathbb{C}^{n \times n}$). Suppose $\lambda = i\beta$ with $\beta \in \mathbb{R}$ is an eigenvalue of A and the largest Jordan block for λ has size $k \geq 1$. Then there is a k -dimensional A -invariant subspace on which*

$$e^{At} = e^{i\beta t} P(t),$$

where $P(t)$ is a matrix polynomial in t of degree at most $k - 1$. In particular, each entry is a finite linear combination of $\cos(\beta t)$ and $\sin(\beta t)$ with polynomial-in- t coefficients. If $k = 1$ (semisimple), the motion is uniformly bounded and purely oscillatory; if $k \geq 2$, the amplitude acquires a polynomial growth factor t^{k-1} .

Proof. Work over $\mathbb{C}$. On the generalized eigenspace of $\lambda = i\beta$, there exists a Jordan block

$$J_k(i\beta) = i\beta I + N, \quad N^k = 0.$$

Using only the exponential series and nilpotency,

$$e^{J_k(i\beta)t} = e^{(i\beta I + N)t} = e^{i\beta t} \sum_{r=0}^{k-1} \frac{(Nt)^r}{r!}.$$

This is $e^{i\beta t}$ times a polynomial in t of degree $\leq k - 1$ with matrix coefficients $N^r/r!$. Returning to real coordinates (taking real and imaginary parts) writes each entry as a finite linear combination of $\cos(\beta t)$ and $\sin(\beta t)$ with polynomial coefficients. If $k = 1$ the sum reduces to I (pure oscillation). If $k \geq 2$, the top term is $(t^{k-1}/(k-1)!) N^{k-1}$, producing polynomial growth t^{k-1} in suitable directions. $\square$

Lemma 7.5 (Real Schur (quasi-triangular) viewpoint). *Over $\mathbb{R}$, let*

$$B = \begin{bmatrix} 0 & -\beta \\ \beta & 0 \end{bmatrix} \Rightarrow e^{Bt} = \begin{bmatrix} \cos(\beta t) & -\sin(\beta t) \\ \sin(\beta t) & \cos(\beta t) \end{bmatrix} =: R(\beta t).$$

(a) *If $T = B$ (a 2×2 real rotation block), then $e^{Tt} = R(\beta t)$: bounded, purely oscillatory.*

(b) *If*

$$T = \begin{bmatrix} B & M \\ 0 & B \end{bmatrix} \quad (\text{a length-2 chain of identical rotation blocks}),$$

then for all $t \geq 0$,

$$e^{Tt} = \begin{bmatrix} e^{Bt} & \int_0^t e^{B(t-s)} M e^{Bs} ds \\ 0 & e^{Bt} \end{bmatrix}.$$

Since $e^{B(t-s)} e^{Bs} = e^{Bt}$ (the same B commutes with itself), the integral collapses to

$$\int_0^t e^{Bt} M ds = t e^{Bt} M,$$

so

$$e^{Tt} = \begin{bmatrix} R(\beta t) & \boxed{t R(\beta t) M} \\ 0 & R(\beta t) \end{bmatrix},$$

i.e. a polynomial factor t multiplying an oscillatory rotation. For longer chains of identical B 's coupled along the superdiagonal, iterating the Volterra formula yields higher polynomial factors $t^r / r!$ multiplying $R(\beta t)$.

Corollary 7.6. *If all Jordan blocks on the imaginary axis have size 1 (semisimple), then $\|e^{At}\|$ is bounded on $\mathbb{R}$ and dynamics are purely oscillatory on those invariant subspaces. If some block has size $k \geq 2$, then along a suitable vector v ,*

$$\|e^{At}v\| \asymp t^{k-1} \quad \text{times a bounded oscillatory factor } (\cos \beta t, \sin \beta t),$$

hence polynomial growth (“polynomial $\times$ oscillatory”).

8. WORKED EXAMPLES

Example 1 (diagonalizable & triangular). Let

$$A = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}.$$

Using the upper-triangular shortcut (with $a = 2$, $d = 3$, $b = 1$) gives

$$e^{At} = \begin{bmatrix} e^{2t} & \frac{e^{2t} - e^{3t}}{2 - 3} \\ 0 & e^{3t} \end{bmatrix} = \begin{bmatrix} e^{2t} & e^{3t} - e^{2t} \\ 0 & e^{3t} \end{bmatrix}.$$

With $x(0) = (0, 1)^\top$: $x(t) = (e^{3t} - e^{2t}, e^{3t})^\top$.

Example 2 (Jordan block, repeated eigenvalue).

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = J_2(1), \quad N = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad N^2 = 0.$$

Then

$$e^{At} = e^t(I + Nt) = e^t \begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix}.$$

So $x(t) = e^t \begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix} x_0$, which shows the characteristic $t e^{\lambda t}$ growth when A is not diagonalizable.

Example 3 (pure rotation).

$$A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}, \quad \text{tr } A = 0, \quad \det A = 1.$$

Using the 2×2 formula with $\mu = 0$, $\beta = 1$,

$$e^{At} = \begin{bmatrix} \cos t & -\sin t \\ \sin t & \cos t \end{bmatrix}.$$

All solutions are rotations at unit speed; the origin is a center.

Example 4 (forced system with constant input).

$$A = \begin{bmatrix} -1 & 2 \\ -2 & -1 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad x(0) = x_0.$$

Here $\operatorname{tr} A = -2$, $\det A = 5 \Rightarrow \mu = -1$, $\beta = 2$. The 2×2 formula gives

$$e^{At} = e^{-t} \left(\cos(2t) I + \frac{\sin(2t)}{2} (A + I) \right) = e^{-t} \begin{bmatrix} \cos(2t) & \sin(2t) \\ -\sin(2t) & \cos(2t) \end{bmatrix}.$$

By variation of constants ($t_0 = 0$),

$$x(t) = e^{At} x_0 + \int_0^t e^{A(t-s)} b \, ds.$$

The integral can be evaluated explicitly, or recognize the equilibrium $x^* = -A^{-1}b$. Since

$$A^{-1} = \frac{1}{5} \begin{bmatrix} -1 & -2 \\ 2 & -1 \end{bmatrix}, \quad x^* = -A^{-1}b = \begin{bmatrix} 1/5 \\ -2/5 \end{bmatrix},$$

we get the clean form

$$x(t) = x^* + e^{At} (x_0 - x^*).$$

Because $\Re(\lambda) = -1 < 0$, solutions spiral into x^* as $t \rightarrow \infty$.

9. REAL VS. COMPLEX SOLUTIONS (HOW TO KEEP THINGS REAL)

Even if A has complex eigenpairs, e^{At} is real whenever A is real. When you diagonalize over $\mathbb{C}$, conjugate pairs combine to yield real sine–cosine expressions (as in Example 4).

10. WHAT IF $f(t)$ IS NOT CONSTANT?

You still use

$$x(t) = e^{A(t-t_0)} x_0 + \int_{t_0}^t e^{A(t-s)} f(s) \, ds.$$

- If f is a polynomial, exponential, or sinusoid, the integral can often be done by hand (integration by parts or Laplace transform).
- For piecewise inputs, split the integral over intervals.
- For general f , this is a *convolution* with kernel e^{At} .

11. QUALITATIVE BEHAVIOR IN THE PLANE (TRACE–DETERMINANT TOOL)

For $A \in \mathbb{R}^{2 \times 2}$, let $\tau = \operatorname{tr} A$, $\Delta = \det A$, and $D = \tau^2 - 4\Delta$.

- **Saddle:** $\Delta < 0$.
- **Node:** $\Delta > 0$, $D \geq 0$ (stable if $\tau < 0$, unstable if $\tau > 0$).
- **Spiral (focus):** $\Delta > 0$, $D < 0$ (stable if $\tau < 0$).
- **Center:** $\tau = 0$, $\Delta > 0$, $D < 0$.
- **Degenerate:** $D = 0$ (repeated eigenvalue); if not diagonalizable, solutions include $te^{\lambda t}$ terms.

12. COMMON PITFALLS AND TIPS

- Do not overuse eigen-decomposition numerically. For hand calculations it is fine; for software, rely on standard expm routines (Padé + scaling/squaring).
- **Defective matrices:** expect polynomial factors $t, t^2, \dots$ multiplying $e^{\lambda t}$.
- **Forcing to an equilibrium:** with constant b , check $x^* = -A^{-1}b$. Writing $x(t) = x^* + e^{At}(x_0 - x^*)$ shows convergence when $\Re \lambda < 0$.
- **Commutation matters:** $e^{(A+B)t} = e^{At}e^{Bt}$ only if $AB = BA$.
- **Determinant growth:** $\det(e^{At}) = e^{(\operatorname{tr} A)t}$ is a quick algebra check.

APPENDIX A. LEIBNIZ RULE (GENERAL BACKGROUND)

Theorem A.1 (Leibniz rule with variable limits). *Let $I \subset \mathbb{R}$ be an open interval. Let $a, b : I \rightarrow \mathbb{R}$ be C^1 functions and let $K : I \times [\alpha, \beta] \rightarrow \mathbb{R}$ be continuous with a continuous partial derivative $\partial_t K$ on $I \times [\alpha, \beta]$, where*

$$\alpha := \inf_{t \in I} \min\{a(t), b(t)\}, \quad \beta := \sup_{t \in I} \max\{a(t), b(t)\}.$$

Define

$$F(t) := \int_{a(t)}^{b(t)} K(t, s) ds \quad (t \in I).$$

Then F is C^1 on I and

$$F'(t) = K(t, b(t)) b'(t) - K(t, a(t)) a'(t) + \int_{a(t)}^{b(t)} \partial_t K(t, s) ds.$$

Proof. Fix $t \in I$ and h small with $t, t+h \in I$. Write

$$F(t+h) - F(t) = \int_{a(t+h)}^{b(t+h)} K(t+h, s) ds - \int_{a(t)}^{b(t)} K(t, s) ds = I_1(h) + I_2(h) - I_3(h),$$

where

$$I_1(h) := \int_{a(t)}^{b(t)} [K(t+h, s) - K(t, s)] ds, \quad I_2(h) := \int_{b(t)}^{b(t+h)} K(t+h, s) ds, \quad I_3(h) := \int_{a(t)}^{a(t+h)} K(t+h, s) ds.$$

We compute the limit of each term divided by h .

Term I_1 : differentiate under the integral. By the mean value form of the fundamental theorem of calculus and continuity of $\partial_t K$,

$$\frac{K(t+h, s) - K(t, s)}{h} \xrightarrow{h \rightarrow 0} \partial_t K(t, s)$$

for each s , and the family is uniformly dominated on $[\alpha, \beta]$ because $\sup_{(t', s) \in J \times [\alpha, \beta]} |\partial_t K(t', s)| < \infty$ on a small compact $J \ni t$. By dominated convergence,

$$\frac{I_1(h)}{h} \xrightarrow{h \rightarrow 0} \int_{a(t)}^{b(t)} \partial_t K(t, s) ds.$$

Term I_2 : moving upper limit. Use the change of variables $s = b(t) + \theta(b(t+h) - b(t))$:

$$I_2(h) = (b(t+h) - b(t)) \int_0^1 K(t+h, b(t) + \theta(b(t+h) - b(t))) d\theta.$$

Divide by h :

$$\frac{I_2(h)}{h} = \frac{b(t+h) - b(t)}{h} \int_0^1 K(t+h, b(t) + \theta(b(t+h) - b(t))) d\theta.$$

As $h \rightarrow 0$, the first factor $\rightarrow b'(t)$ and the integrand converges uniformly to $K(t, b(t))$ by continuity of K ; hence

$$\frac{I_2(h)}{h} \xrightarrow{h \rightarrow 0} K(t, b(t)) b'(t).$$

(Equivalently, one may invoke the integral mean value theorem: $I_2(h) = K(t+h, \xi_h)(b(t+h) - b(t))$ for some ξ_h between $b(t)$ and $b(t+h)$, then pass to the limit.)

Term I_3 : moving lower limit. A completely analogous argument (or the same change of variables) yields

$$\frac{I_3(h)}{h} \xrightarrow{h \rightarrow 0} K(t, a(t)) a'(t).$$

Summing the three limits gives the claimed formula for $F'(t)$. Since t was arbitrary and the ingredients are continuous, F' is continuous on I . $\square$

Remark A.2 (Specialization to Duhamel). If $a(t) \equiv t_0$ (constant) and $b(t) = t$, and $K(t, s) = e^{A(t-s)}f(s)$ with $f \in L^1_{\text{loc}}$, the same proof works on each compact time window because $t \mapsto e^{A(t-s)}$ and $t \mapsto Ae^{A(t-s)}$ are continuous and uniformly bounded there. The rule reduces to

$$\frac{d}{dt} \left(\int_{t_0}^t e^{A(t-s)} f(s) ds \right) = e^{A(t-t)} f(t) + \int_{t_0}^t Ae^{A(t-s)} f(s) ds = f(t) + \int_{t_0}^t Ae^{A(t-s)} f(s) ds.$$